

Hackathon Winners

Random Forest

↳ Ensemble Learning

Bagging $\Rightarrow$ BOOSTSTRAPPING + AGGREGATION

D.T $\Rightarrow$ highly Sensitive Training $\Rightarrow$ one perspective.

BOOT STRAPPING

↳ Randomness

↳ ① rows - 2
↳ ② columns - 3



Bootstrapping $\Rightarrow$ multiple unique subsets of data are generated from the original training dataset by

in RF $\Rightarrow$ Rows are sampled so Row is Bootstrapped.

$\Rightarrow$ Bootstrapped Dataset $\Rightarrow$ Build Multiple Trees.

AGGREGATE $\Rightarrow$ RESULTS!!

How often two rows
land in same leaf.

Proximity Matrix

BOOSTING

TREE 1
INDEPENDENTLY

TREE 2
IND

TREE 3
IND.

LAB A Same fields.

⇒ Research 1 / Research 2 / 3..

IND. Projects

IND. Projects

IND. Projects

New Juniors → IND. projects

They can span Multiple application Ideas / Multiple perspectives

BAGGING!

Lab B

Research 1 / Research 2

Proj 1 / Proj 2

Multiple Pen in Depth

Juniors learn from "Mistakes" of Seniors

Learning from Mistakes

BOOSTING ⇒ SEQUENTIALLY

→ Several Weak Models

Strong Model.

Each Model tries to correct the mistakes of previous model.

... → Weak Models

Weak Model 1 $\rightarrow$ Weak Model 2 $\leftarrow$ (fix mistakes of weak model 1)
 (fix mistakes of prev. models)

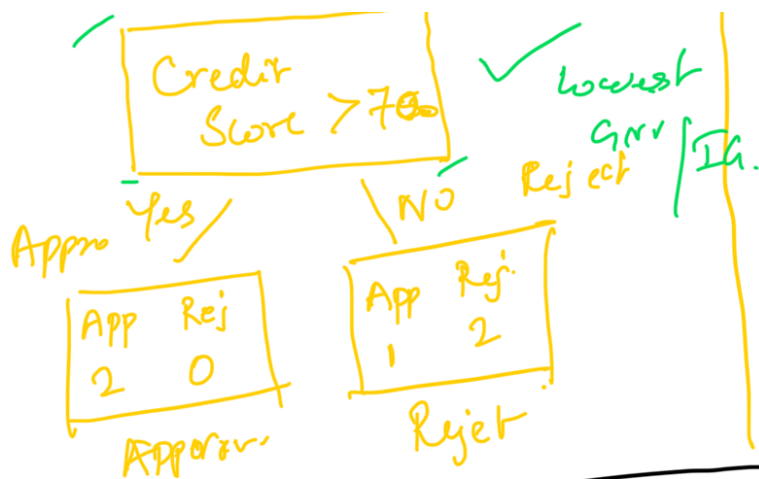
Loan ID	Credit Score	Income (\$k)	Debt	Loan Status
1	750	70	High	✓ ✓
2	680	55	low	✓ ✗
3	620	65	High	✗ ✗
4	590	30	High	✗ ✗
5	720	70	low	✓ ✓

$\Rightarrow$ Keep track of rows that are harder to predict.

I am making Models Sequentially correct mistakes
 $\hookrightarrow$ I am going to increase their representation more & more.

Loan ID	True Answer	Starting Weight
1	✓	$1/5 = 0.2$
2	✓	$1/5 = 0.2$
3	✗	$1/5 = 0.2$
4	✗	$1/5 = 0.2$
5	✓	$1/5 = 0.2$

① Weak learners / one column.



True	Pred.
✓	✓
✓	x
x	x
x	x
✓	✓

1/5

2/4 = 0.5

Weak learn. Error = 0.20 ⇒ 1/5 = 0.20

Amount of Sag $\alpha = \frac{1}{2} \ln \left(\frac{1 - \text{error}}{\text{error}} \right) = \frac{1}{2} \ln \left(\frac{0.8}{0.2} \right) \approx 0.693$

non zero.

Wrong

$e^{0.693} \approx 2$

Correct Row Weigh.

$e^{-0.693} \approx 0.5$

F_1

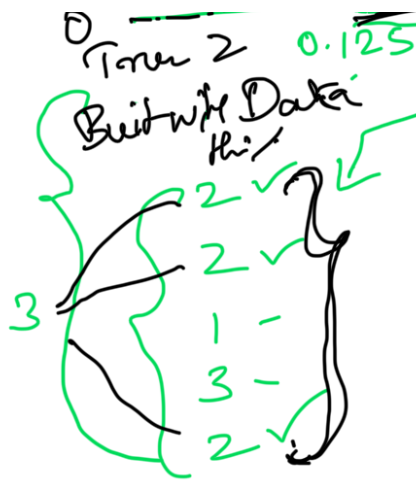
loan ID	Pred Correct.	Old	Multin	New Weigh.
1	Yes	0.2	$0.2 \times 0.5 = 0.10$	0.125
2	No	0.2	$0.2 \times 2 = 0.40$	0.5
3	Yes	0.2	$0.2 \times 0.5 = 0.10$	0.125
4	Yes	0.2	$0.2 \times 0.5 = 0.10$	0.125
5	Yes	0.2	$0.2 \times 0.3 = 0.10$	0.12

Round(0.1) 0.100

0.10 / 0.80 = 0.125

0.4 / 0.80 = 0.5

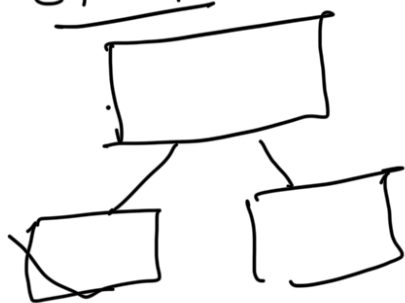
0.72 0.80



① BOOSTING

Ada Boost → BOOSTING WHEN SEQUENTIAL MISTAKE

STUMPS



① An Army of Weak learners (Trees with just 1 column) ^{CORR. CTIO}

② TABULAR (WEIGHT AGES) (STUMPS) for each Row

Each column ∈ their I.G. which ever column ↑ IG that will be first weak learner.

③ Weak learner → 1 → Predictions. (HIGHEST IG Col)

④ Predictions → none
 Correct predictions (IG root | ENTIRE DATA is enough to predict some)

WRONG PREDICTIONS (INCREASES WEIGHTS OF THESE ROWS) TRIES TO INCREASE REPRESENTATION

NEXT TREE

Repeated Mistake

(5) Focus by Weak learner 1 $\rightarrow$ TREE 2
 Solve Weakly
 Predict more from Strong 1
 rather than all rows at once $\leftarrow$

	AdaBoost	RF
Focus on		
Mistakes	Build Sequence $\checkmark$	IND. $\times$
Sensitivity to Noisy	More	Less
Parallelization	Easier	Harder
	Correct Mistakes	Average may free

Is Input $\geq$ Split	Yes	No	Tree	Index	Actual	Pred ₂	Correct
Yes	$\rightarrow$		1	70	$\checkmark$	$\checkmark$	Yes
No		$\rightarrow$	2	55	$\checkmark$	$\checkmark$	Yes $\Rightarrow$
Reject		$\rightarrow$	3	65	$\times$	$\checkmark$	Wrong
		$\rightarrow$	4	30	$\times$	$\times$	Yes
		$\rightarrow$	5	40	$\checkmark$	$\times$	Wrong

Current New 2nd Tree

0.125 0.08

0.500 0.33

$$E_{r_2} = 0.125 + 0.125 = 0.25$$

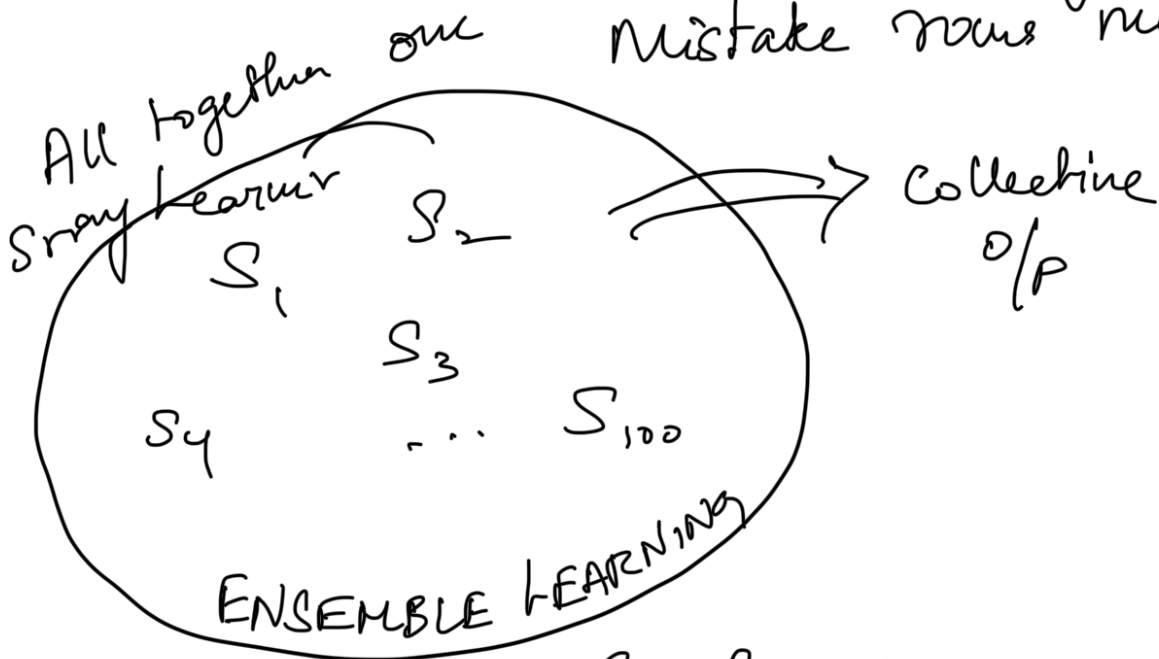
$$\ln(1/0.25) \approx 1.39$$

100/60-70

$$\alpha_2 = \frac{1}{2} \ln \left(\frac{1 - 0.25}{0.25} \right) = \frac{1}{2} \ln 3$$
 No Change in α

0.125 0.9
0.125 0.8
0.125 0.25
1.0

Data But TREES
Will Try to solve
Mistake runs more!

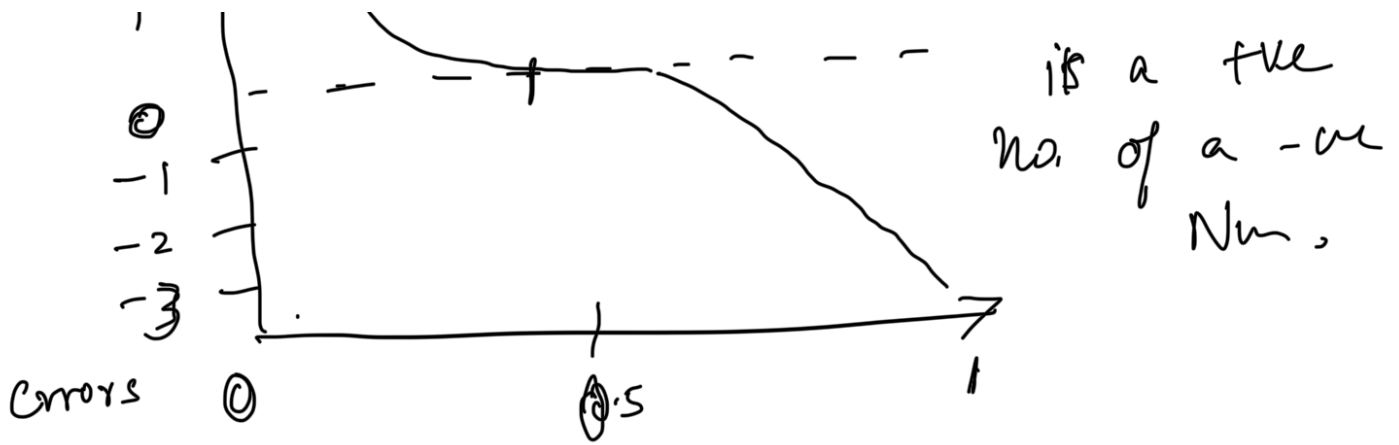


$\Rightarrow$ / Examples are Simple X
 10/1000 / Examples are genuine difficult ✓

STUMPS	rule	Error	Amount
1	Cv.S ≥ 700	0.20	0.69 Say -
2	Income $\geq 50k$	0.25	0.54
3	Debt = low	0.083	1.19

3
 2
 1

Amount of Say doesn't have to be 0-1



$$f_{\text{final}} = (\alpha_1 h_1 + \alpha_2 h_2 + \alpha_3 h_3)$$

$f_1 = \text{Approve} \rightarrow$

$-1 = \text{Reject}$

Approve

Row 6

<u>STUMP 1</u>	- Approve	<u>STUMP 2</u>	<u>STUMP 3</u>
$\Rightarrow +0.693$		$+0.549$	-1.199

$$(0.693 \times f_1) + (0.549 \times f_1) + (1.199 \times -1)$$

$$= +0.043 = \text{Approved}$$

$$\frac{1}{2} \ln \left(\frac{\text{error}}{\text{error}}$$

① Ada-Boost $\Rightarrow$ Adaptive Boosting

1. Why is it called "Adaptive"?

$\rightarrow$ Samples misclassified get more weight

2.) What is the weak learner here?

- 1 $\Rightarrow$ SIMPLE
- 3.) Can Ada Boost use deeper Trees?
 $\Rightarrow$ Yes. But will cause over fitting
- 4.) How does it focus on which sample goes into stumps next?
 $\Rightarrow$ Exponential increase/decreases
- 5.) How are final predictions taken
 $\Rightarrow \text{Sign}(\sum_{t=1}^T \alpha_t h_t(x))$
- 6.) Is Ada Boost sensitive to outliers?
 $\Rightarrow$ Yes
- 7.) Can it overfit?
 $\Rightarrow$ Yes, -